



BRENT OZAR
UNLIMITED®

Designing Backups in the Cloud

(And yes, this is a performance issue, too)

Deck 5 p1

Backups are a performance issue.

They're my favorite performance test.

You're already doing them.

They're easily repeatable.

Fast backups don't guarantee fast queries,
but if your backups can't keep up,
odds are your queries are screwed, too.



Deck 5 p2

Agenda

How Alex, Billie, and Elliot backed up on-premises

The performance bottleneck for backups

Options to work around it

How VM-based backups are affected

Why we back up: specific risks

Why we check cloud backup & restore performance

Deck 5 p3



Deck 5 p4





Elliot's Enterprise backups

One backup tool for the entire shop

Policy-based retention, encryption

Syncs offsite automatically

DBAs have minimal control or input

Backups & restores typically slow

Deck 5 p5



Billie's Small Business Backups

Most servers use a VM-based backup tool

Their largest SQL Server might do native SQL Server backups written to a file share, and then something else archives those files



Deck 5 p6





Alex the Architect's backups

Alex... doesn't usually think about this.

Alex doesn't *want* to think about this.

Just wants backups to work.

Expects to be able to do point-in-time recovery using a simple wizard.

After all, it's 2023: how hard can backups be?

Deck 5 p7



Let's focus on SQL backups first.

Some companies do VM-level backups.

Tools like Veeam take a snapshot of the entire server, typically every 15 minutes or less frequently, and then restore the entire server.

That does not work well with Availability Groups, mirroring, etc.

I'll come back to those. They have a new problem in the cloud, but I need to explain natives first.

Deck 5 p8



The cloud backup bottleneck



Backup performance is about

Faster backups: shorter maintenance windows

Faster restores:

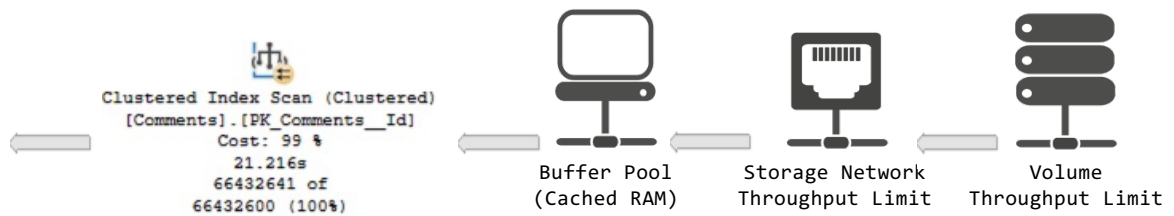
- Shorter Recovery Time Objective
- Easier re-seeding of Availability Groups
- Quicker refreshes of other environments

If you don't have to worry about these, great!
(But most shops do.)



Remember this?

We talked about how execution plans should really keep going to the right:

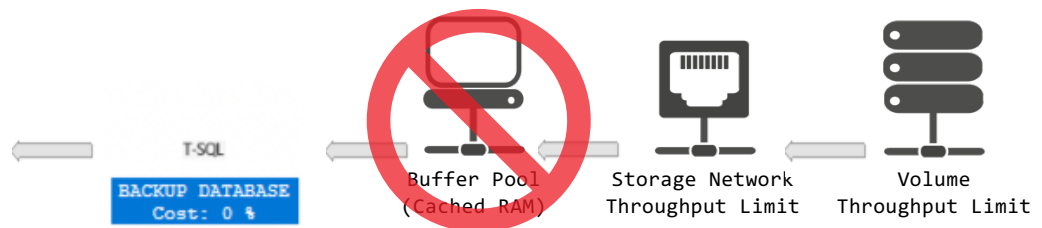


Deck 5 p11



It's true for backups too...

They read from the data files just like queries do, but technically, they don't use cached RAM.



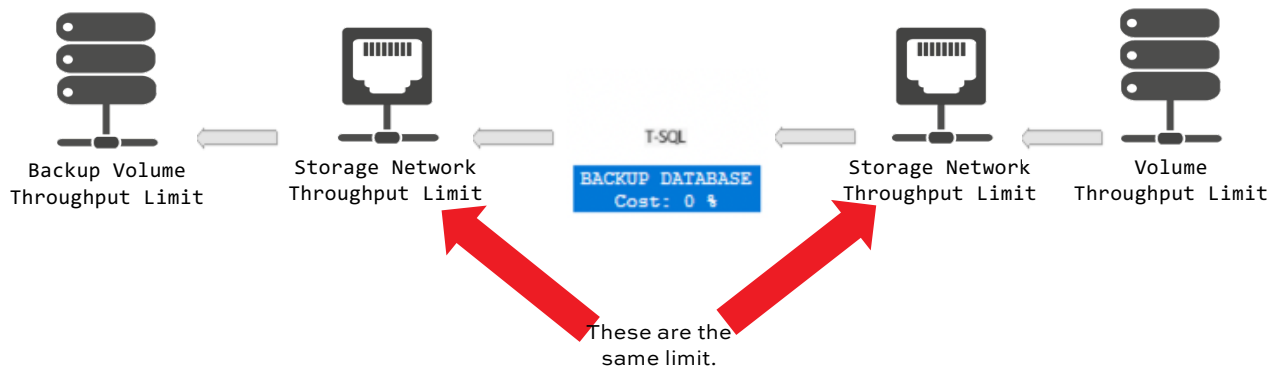
But where do we write backups to?

Deck 5 p12



If you're writing to a volume...

That's attached to the same VM you're backing up,
then you're cutting your own storage throughput.



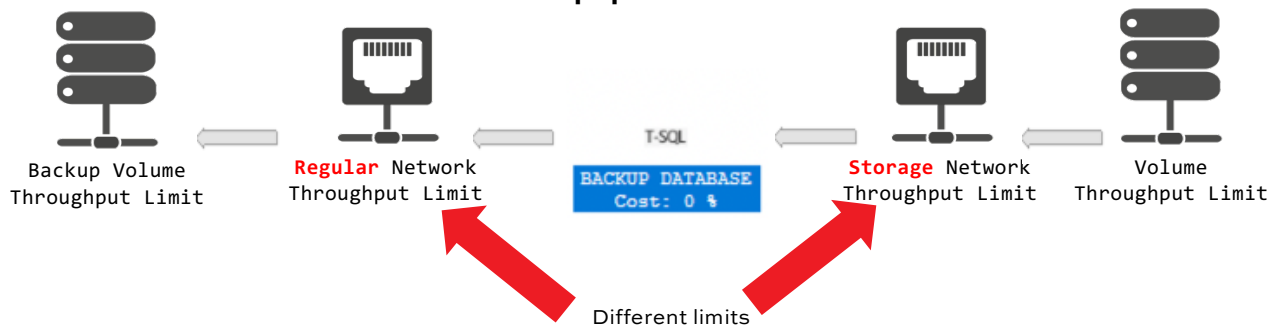
Deck 5 p13



This isn't just the cloud.

But it's *especially* bad in the cloud because the
storage network throughput is so small.

We need to use another pipe for writes instead:



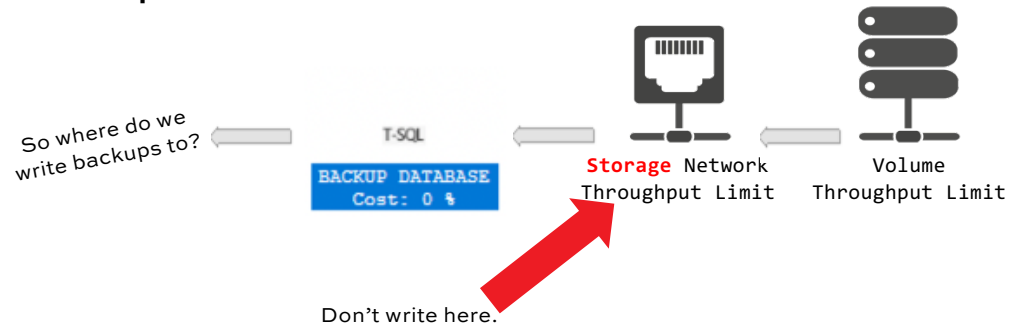
Deck 5 p14



This isn't just the cloud.

But it's *especially* bad in the cloud because the storage network throughput is so small.

We need backups that don't write to this bottleneck.



Deck 5 p15



Option 1:
local SSD

Ephemeral solid state

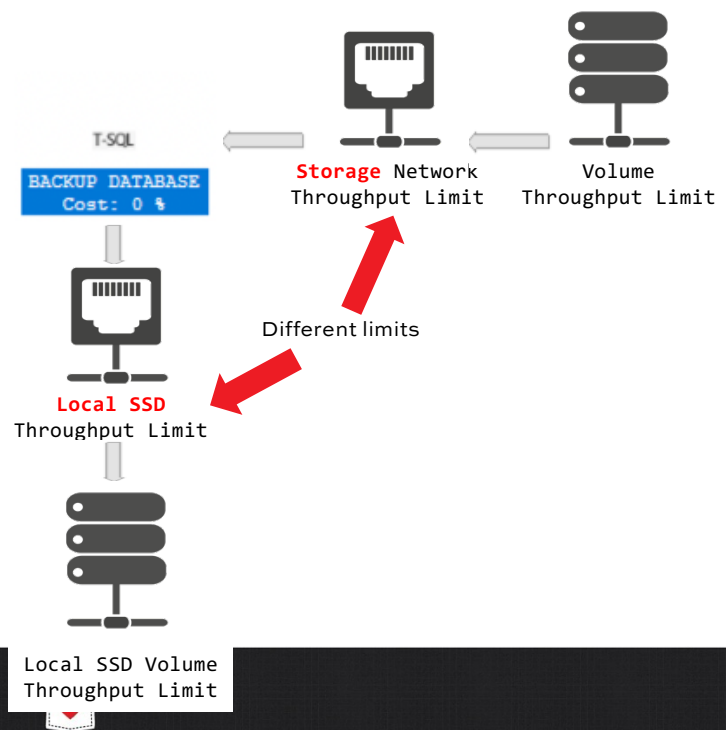
Many (not all) VM types in AWS & Azure come with free, fast local solid state drives.

The bigger the VM, the more space you get.

Size	vCPU	Memory GiB	Temp storage (SSD) GiB	Max data disks	Max temp storage throughput: IOPS / MBps	Max uncached storage throughput: IOPS / MBps	Max burst uncached disk throughput: IOPS/MBp	Max NICs	Network bandwidth
Standard_E2bds_v5	2	16	75	4	9000/125	5500/156	10000/1200	2	10000
Standard_E4bds_v5	4	32	150	8	19000/250	11000/350	20000/1200	2	10000
Standard_E8bds_v5	8	64	300	16	38000/500	22000/625	40000/1200	4	10000
Standard_E16bds_v5	16	128	600	32	75000/1000	44000/1250	64000/2000	8	12500
Standard_E32bds_v5	32	256	1200	32	150000/1250	88000/2500	120000/4000	8	16000

Deck 5 p1

The storage network isn't involved.



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But the bad news...

Backups are only backups if you can read ‘em.

If the VM goes down, you can’t read this backup.

You have to get the backup off the local SSD quickly.

You have to push it across the network,
which can make your RPO & RTO worse.

Deck 5 p19



The part I really hate

“We wrote scripts to sync the files”

If I had a dollar for every time I heard this...

and upon investigation, the scripts weren’t working...

and nobody knew about it until I looked...

I’d have a lot of dollars.

Deck 5 p20





Elliot and I are not fans.

I understand a lot of people are.

Alex the Architect might obsess over really fast backups – because Alex hasn't had to deal with a dead server whose sync scripts failed.

Billie the Small Business Bird would try it, and then regret it later without the full time staff to watch the scripts.

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Option 2:
a network share

Back up over the network.

Don't back up to volumes attached to the server.

Instead, write your backups to:

- A fast shared UNC path hosted on another server (traditional route)
- A backup appliance like Data Domain (traditional on premises, less popular in cloud)
- A file-share-as-a-service
- Blob-store-as-a-service

Deck 5 p23



Option A: File-share-as-a-service

Azure Files, Amazon FSx: managed file shares

 Amazon FSx for NetApp ONTAP Fully managed shared storage built on NetApp's popular ONTAP file system.	 Amazon FSx for OpenZFS Fully managed shared storage built on the popular OpenZFS file system.	 Amazon FSx for Windows File Server Fully managed shared storage built on Windows Server.	 Amazon FSx for Lustre Fully managed shared storage built on the world's most popular high-performance file system.
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Configuring this is easy. It's just an NFS or SMB share:
`\\servername\sharename\mybackupfile.bak`

Deck 5 p24



Option B: Blob-store-as-a-service

Azure Blob Storage, Amazon S3

Not directly accessible as NFS or SMB file shares

Backup app needs to speak the ABS or S3 protocol

	Back up directly to Azure Blob Storage	Back up directly to AWS S3
SQL Server 2012 & newer	Yes	-
SQL Server 2022 & newer	Yes	Yes
3 rd party utilities like Quest Litespeed	Some	Some

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Configuring Option B: Blobs

Writing backups to Azure Blob Storage (2012+):

<https://docs.microsoft.com/en-us/sql/relational-databases/tutorial-sql-server-backup-and-restore-to-azure-blob-storage-service>

Writing backups to Amazon S3 (2022+):

<https://docs.microsoft.com/en-us/sql/relational-databases/backup-restore/sql-server-backup-to-url-s3-compatible-object-storage>

Or use 3rd party products: Idera SQL Safe, Quest Litespeed, Redgate SQL Backup

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The really cool part

DBAs are used to:

- Building jobs to delete old files
- Worrying about whether the backups got copied
- Worrying about whether they made it offsite

But with Amazon S3 & Azure Blob Storage, that can all be automated with policies.

(You'll still worry about it.)

Deck 5 p27



Configuring retention, replication

Amazon:

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lifecycle-mgmt.html>

Azure: <https://docs.microsoft.com/en-us/azure/storage/blobs/immutable-time-based-retention-policy-overview>

Both let you set time policies, sweep old files to cheaper storage, copy to other accounts for safety.

This is a sysadmin function, not a DBA function.

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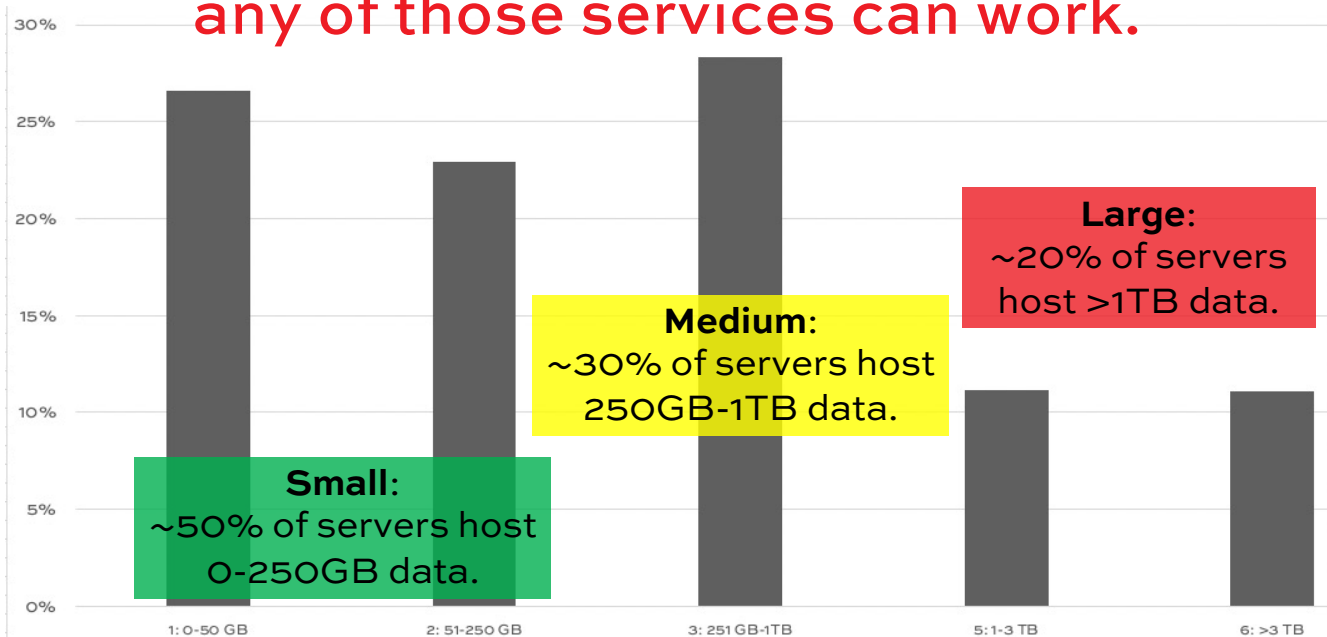
Comparison

	Option B for Blob: Azure Blob Storage & AWS S3	Option A: Azure Files & AWS FSx
Been around for	A really long time	Relatively new
Max share size	Unlimited	64-100 TB
Max throughput	Limited	Limited
Priced by	Space used, IO	Monthly server price, PLUS space used, IO
Easy to replicate globally	Yes	Kinda
Summary	Cheap, reliable (My preference)	More expensive, more things can go wrong

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For small & medium servers,
any of those services can work.



For >1TB databases, it's harder.

Backup & restore speed matters:
performance tune writing to multiple files, even
multiple storage accounts for max throughput.

Max file size for the backup target can matter: for
example, Azure Blob Storage has 200GB file limits:
<https://techcommunity.microsoft.com/t5/sql-server-support-blog/sql-backup-using-azure-file-share/ba-p/3596366>

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Option 3: storage snapshots

On-premises: popular for VLDBs

Because multi-terabyte databases are:

- Usually on shared storage (SANs)
- Usually in failover clustered instances (FCIs)
- Painful to restore quickly,
like to refresh development environments

Deck 5 p33



Storage snapshots: easy option

Storage driver tells Windows, “Freeze your writes”

Windows Volume Shadow Copy Service (VSS)
notifies all running applications to pause writes

SQL Server freezes its writes

Storage makes a note of where data lives now, tells
Windows VSS to thaw writes

All subsequent writes go to different area of storage

Deck 5 p34



Benefits

No matter how big a db is:

- Backups take seconds
- Restores take seconds: easier refreshes
- Offsite replication is easier/faster
(in both directions, too – only changes are sent)

But it does require compatible storage.

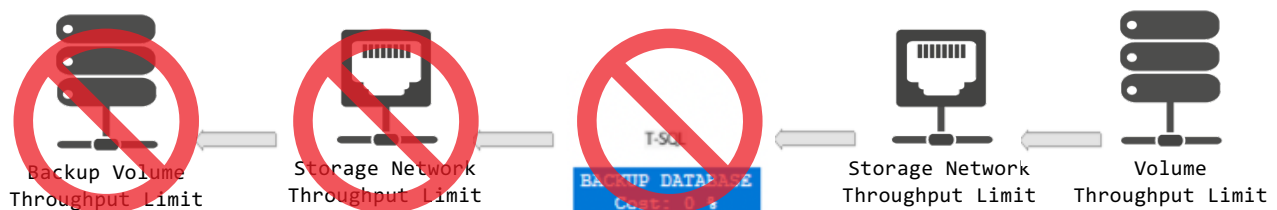
Deck 5 p35



Cool on-premises because...

The backup data doesn't go out through SQL Server.

It doesn't go out through the OS, either.



Deck 5 p36

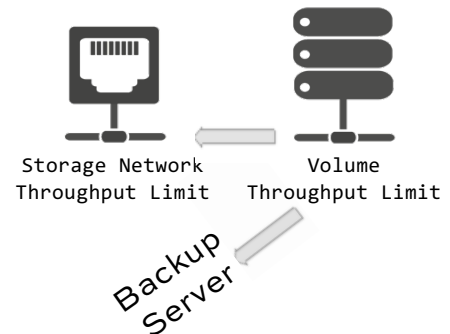


Backup software directly hits storage.

The backup software runs on its own server or VM.

The backup software can read the data when it's convenient.

SQL Server doesn't slow down (much) while the backups are read compared to pulling the entire db through SQL Server's storage pipe.



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Documentation from vendors

AWS:

<https://docs.aws.amazon.com/AWSEC2/latest/WindowsGuide/application-consistent-snapshots.html>

Azure: <https://docs.microsoft.com/en-us/azure/backup/backup-azure-vms-introduction>

SQL Server 2022 added T-SQL to freeze IO, but still needs something on the storage side to actually do the backup part.

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But these are tools. You have to:

Schedule the jobs

Script the restores (it's not in SSMS)

Define storage policies to:

- Assign snapshots the right cost tags
- Sync snapshots between locations
- Delete old snapshots

Deck 5 p39



Drawback: ~35 databases max

<https://docs.microsoft.com/en-US/troubleshoot/sql/admin/error-when-you-create-a-snapshot>

The screenshot shows a Microsoft Docs page. The header includes the Microsoft logo and navigation links: Docs, Documentation, Learn, Certifications, Q&A, Code Samples, Shows, and Events. A search bar is present with the text 'Filter by title'. Below the search bar is a list of search results, including 'DBCC CHECK', '8967 and 8921 errors when running DBCC CHECKDB', 'A nonrecoverable I/O error when you use Backup to URL', 'A TDE-enabled database may not recover', 'Access violations and memory dump files', 'Activation tasks run over limit', and 'Agent job may fail with error 65535'. The main article title is 'Error when you create a snapshot backup of many databases at the same time in SQL Server'. Below the title, it says 'Article • 01/25/2022 • 2 minutes to read • 3 contributors'. The article content begins with 'This article helps you work around for the problem (ERROR Selected writer 'Microsoft Writer (Service State)' is in failed state message) that when you create a snapshot backup of many databases at the same time in SQL Server.'

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Workaround

To work around this problem, create a snapshot backup of fewer databases at the same time.

More information

We recommend that you create a snapshot backup of fewer than 35 databases at the same time.

Got 35+ databases on the same volume?

You can't pick and choose which ones get frozen: all databases on a volume are always frozen, every time.

Deck 5 p41



Option 4:
VM snapshots

How storage* snapshots work

(* we're not talking about VMs yet)

Storage driver tells Windows, “Freeze your writes”

Windows Volume Shadow Copy Service (VSS)
notifies all running applications to pause writes

SQL Server freezes its writes

Storage makes a note of where data lives now, tells
Windows VSS to thaw writes

All subsequent writes go to different area of storage

Deck 5 p43

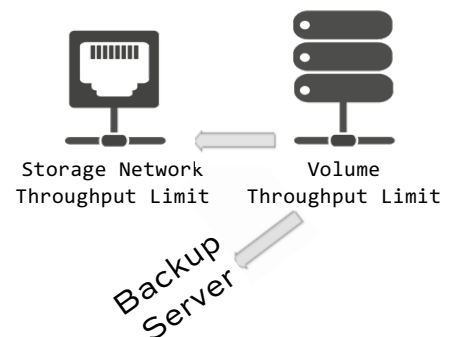


On-premises, this works great.

The backup software runs on its own server or VM.

The backup server pulls the data
directly from the storage volume.

The SQL Server doesn't slow down
while the backup data is read.



But the cloud can be different.

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How some VM backup apps work

Some backup apps don't access storage directly.

They use Windows' built-in VSS snapshots
and keep the snapshot data on the VM itself.

To get the data out of the VM, they have to:

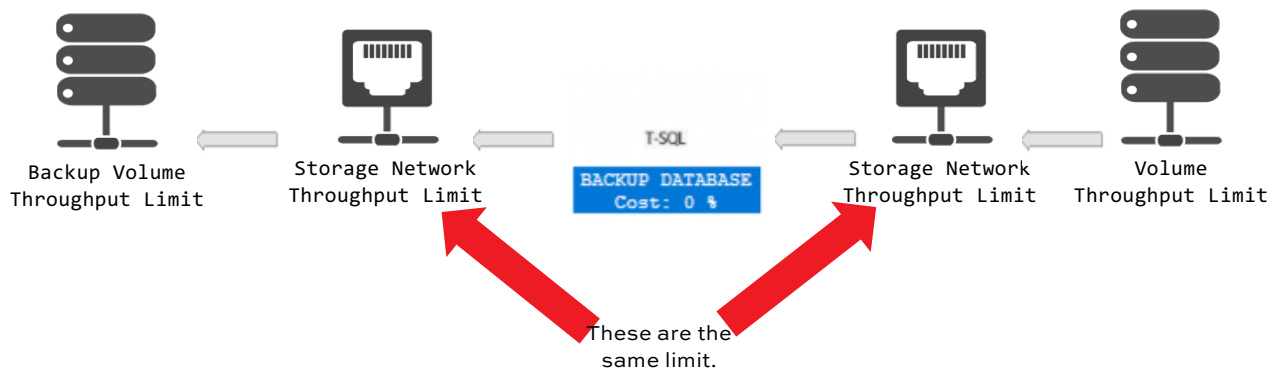
- Access the same underlying drives as the VM
- Read the data through the VM's network
- Write the data out through the VM's network, too
- Slow down processing on the VM

Deck 5 p45



THAT'S THE PROBLEM.

This is the same reason we can't do native SQL
Server backups to local drives: performance sucks.



Deck 5 p46



Questions to ask backup vendors

Can you take Amazon EBS or Azure Blob snapshots?

Can other VMs access those EBS or Blob snapshots directly, without going through the backed-up SQL Server?

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How to prove it

1. Take a backup of your largest database. Time it.
It should only take seconds.
2. Immediately after the backup finishes,
restore that exact backup to a dev or test server.
That should only take seconds too.

If either of those operations takes >15 seconds,
or if the backup software makes you wait between
steps 1 and 2, that's a problem.

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Recap of 4 common backup options

1. Write to local SSD, copy the backup file off later
2. Write backup directly to a network share
3. Cloud provider's EBS & Blob Storage snapshots
4. Backup apps doing VM-level snapshots

Before we pick a winner, what are backups for?

Deck 5 p49



Why we back up

Risks we've always had

- Servers fail, drop offline
- Databases get corrupted
- People delete rows, drop tables
- Deployments fail

So for SQL Server, we've always:

- Built HA/DR with clusters, log shipping
- Taken log backups frequently
- Copied/moved backups to other locations

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But there are new cloud-specific risks.

Deck 5 p52



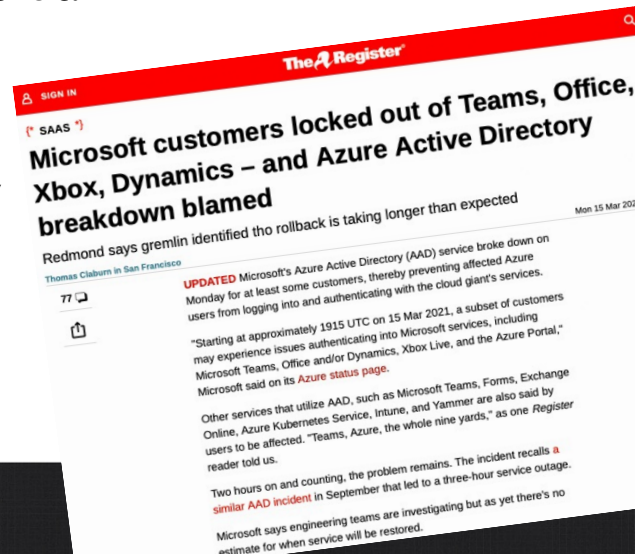
“The cloud might go down.”

I’m not really worried about that.

The *entire* provider rarely goes down entirely.

When it does, we can’t really work around that.

Multi-cloud solutions are crazy complex & expensive.



Deck 5 p53



One region might go down.

This happens much more frequently.

Even more common:

- Some services are down in your primary region
- But other services are up

So do you fail everything over?
Which services need DR?



Deck 5 p54



Cloud & region failures: rare.

Before you dismiss them as unrealistic,
let's talk about a different kind of failure.

It's more common than cloud or region failure.

But for me, it's scarier.

To understand what it is, you have to start by
understanding a good cloud best practice...

Deck 5 p55



Using multiple accounts is good.

Cloud providers straight-up encourage it.

<https://docs.aws.amazon.com/whitepapers/latest/organizing-your-aws-environment/benefits-of-using-multiple-aws-accounts.html>

The screenshot shows the AWS documentation page for the whitepaper 'Benefits of using multiple AWS accounts'. The page has a dark header with the AWS logo and a search bar. Below the header is a breadcrumb trail: 'AWS > Documentation > AWS Whitepapers > AWS Whitepaper'. The main content area is divided into two columns. The left column contains a sidebar with the title 'Organizing Your AWS Environment Using Multiple Accounts' and a list of links: 'Organizing Your AWS Environment' and 'Introduction'. The right column features the title 'Benefits of using multiple AWS accounts' with links for 'PDF' and 'RSS'. Below the title, the text reads: 'As you adopt AWS, we recommend that you determine how your business, governance, security, and operational requirements can be met in AWS. Use of multiple AWS accounts plays an important role in how you meet those requirements.'

Group workloads based on business purpose and ownership

You can group workloads with a common business purpose in distinct accounts. As a result, you can align the ownership and decision making with those accounts and avoid dependencies and conflicts with how workloads in other accounts are secured and managed.

Different business units or product teams may have different processes. Depending on your overall business model, you might choose to isolate distinct business units or subsidiaries in different accounts. Isolation of business units can help them operate with greater decentralized control, but still provides the ability for you to provide overarching guardrails. This approach might also ease divestment of those units over time.

Guardrails are governance rules for security, operations, and compliance that you can define and apply either across your AWS environment or to specific groups of accounts. Guardrails help protect your users from making choices that aren't aligned with your overall requirements.

If you acquire a business that is already operating in AWS, you can move the associated accounts intact into your existing organization. This movement of accounts can be an initial step toward integrating acquired services into your standard account structure.

Apply distinct security controls by environment

Workloads often have distinct security profiles that require separate control policies and mechanisms to support them. For example, it's common to apply different security and operational policies for the non-production and production environments of a given workload. By using separate accounts for the non-production and production environments, by default, the resources and data that make up a workload environment are separated from other environments and workloads.

Constrain access to sensitive data

When you limit sensitive data stores to an account that is built to manage it, you can more easily constrain the number of people and processes that can access and manage the data store. This approach simplifies the process of achieving least privilege access. Limiting access at the coarse-grained level of an account helps contain exposure to highly sensitive data.

For example, designating a set of accounts to house publicly accessible Amazon S3 buckets would enable you to implement policies for all your other accounts to expressly forbid making S3 buckets publicly available.

Deck 5 p57



Azure? Similar guidance.

<https://docs.microsoft.com/en-us/azure/cloud-adoption-framework/ready/landing-zone/design-area/resource-org-subscriptions>

The screenshot shows the Microsoft Azure documentation website. The top navigation bar includes the Microsoft logo and links to Docs, Documentation, Learn, Certifications, Q&A, Code Samples, Shows, and Events. Below this, the 'Azure' section is expanded, showing links to Product documentation, Architecture, Learn Azure, Develop, and Resources. The left sidebar contains a search bar and a list of topics under 'Cloud Adoption Framework for Azure', including 'About the Framework', 'What's new', 'Scenarios', 'Adoption journeys', 'Strategy', 'Plan', 'Ready', and 'Azure landing zones'. The main content area displays the article 'Subscription considerations and recommendations', dated 06/29/2022, with a 7-minute read time and 9 contributors. The article text states that subscriptions are a unit of management, billing, and scale within Azure, and lists four critical factors for design: environment type, ownership and governance model, organizational structure, and application portfolios.

Organization and governance design considerations

- Subscriptions serve as boundaries for Azure Policy assignments.
 - For example, secure workloads like Payment Card Industry (PCI) workloads typically require other policies in order to achieve compliance. Instead of using a management group to collate workloads that require PCI compliance, you can achieve the same isolation with a subscription, without having too many management groups with a few subscriptions.
 - If you need to group together many subscriptions of the same workload archetype, create them under a management group.
- Subscriptions serve as a scale unit so component workloads can scale within platform [subscription limits](#). Make sure you consider subscription resource limits as you design your workloads.
- Subscriptions provide a management boundary for governance and isolation that clearly separates concerns.
- Create separate platform subscriptions for management (monitoring), connectivity, and identity when they're required.
 - Establish a dedicated management subscription in your platform management group to support global management capabilities like Azure Monitor Log Analytics workspaces and Azure Automation runbooks.
 - Establish a dedicated identity subscription in your platform management group to host Windows Server Active Directory domain controllers when needed.
 - Establish a dedicated connectivity subscription in your platform management group to host an Azure Virtual WAN hub, private Domain Name System (DNS), ExpressRoute circuit, and other networking resources. A dedicated subscription ensures that all your foundation network resources are billed together and isolated from other workloads.
 - Use subscriptions as a democratized unit of management aligned with your business needs and priorities.

This mitigates risk.

The good:

If someone hacks your internal, user, development, etc. subscription, they may not be able to access resources in your other subscriptions.

The bad:

That only works if you set up security between 'em.

The ugly:

More subscriptions = more moving parts.



True story #1

Client had dozens of separate subscriptions for:

- Production PCI-compliant sensitive services
- Production non-PCI services
- QA, dev, DR, etc for each of the above
- End user networks for employees
- End user networks for contractors

And IT staff were responsible for creating & dropping subscriptions as necessary.

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You guessed it

Someone deleted the wrong one.

They copy/pasted account IDs between windows, didn't notice that a copy didn't take.

They pasted a production subscription ID into an automation script to destroy a subscription's resources and then close the account.

Within minutes, their servers were down.

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You can reactivate it, but...

After reactivation

After your subscription is reactivated, there might be a delay in creating or managing resources. If the delay exceeds 30 minutes, contact [Azure Billing Support](#) for assistance. Most Azure resources automatically resume and don't require any action. However, we recommend that you check your Azure service resources and restart any that don't resume automatically.

You still have to:

- Get Windows up everywhere
- Get cluster services running
- Bring SQL Server back online
- Make sure AG replication is working

Deck 5 p63



True story #2: expired card

This happens a lot. Example:

<https://news.ycombinator.com/item?id=32547912>

Hacker News new | threads | past | comments | ask | show | jobs | submit BrentOzar (3088) | logout

▲ Tell HN: Google Cloud suspended our production projects at 1am on Saturday
1313 points by 7tech 14 days ago | hide | past | favorite | 508 comments

TLDR; never use google cloud systems for production.

Google cloud suspended all our projects due to the billing issue in their system they had.

Despite reassurances "your account will not be suspended" while communicating with billing support, all the projects were suspended at 1am on Saturday.

All the account payments were made and the billing cards are valid. There are no outstanding bills.

Never use GCP for production.

---- Edit: full story by request, long read: ----

Previous month billing didn't went through. Not sure if this was due to the billing outage google had (<https://status.cloud.google.com/products/oLCqDYkE9NFWQVgctQTL/history>) or financial transaction issue, however we went ahead and made a manual payment covering all the outstanding amount + extra. Despite the payment made, about a week+ later we suddenly started receiving threatening emails "Your Projects are at risk of suspension". Edited and updated the billing cards. Opened a billing support request clearly mentioning this is a production environment and all the bills are paid. They were "investigating" the issue and assured the project will not be suspended.

Client example

Cloud account was set up on company AmEx

Company owner's personal account went overdue

AmEx limited both the owner's card,
AND the company's AmEx card

Cloud vendor couldn't charge the company AmEx

Billing emails were ignored

Cloud account shut down, services went offline

Deck 5 p65



True story #3: retention snafu

Client's sysadmin team set up retention policies

In SQL Server backup folder, policy was:
files were kept for 60 days, then deleted

Later, a sysadmin applied the wrong policy to the
backup folder, deleting backup files after 1 week

DBAs didn't find out until they needed a restore

Companies rarely monitor their backup folders

Deck 5 p66



So let's zoom out.

Why we back up

Servers fail, drop offline

Databases get corrupted

People delete rows, drop tables

Deployments fail

Cloud vendor accounts go offline

Cloud availability zones / regions fail

Cloud vendors entirely fail (rarely)



A good backup strategy survives

Disk failures

VM failures

Availability zone / region failures

Account failures

Cloud vendor failures (maybe)

Deck 5 p69



This slide is opinionated.

	Back up to local SSD, then copy to network	Back up to network share	Storage snapshots (either SQL or VM)
Disk failures	Sometimes	Yes	Sometimes
VM failures	Not quickly	Yes	Sometimes
AZ/region failures	Not quickly	If configured well	If configured well
Account failures	Not quickly	If configured well	If configured well

Let's talk through each of the boxes for my opinions.

Deck 5 p70



I love backing up to S3 & Blobs.

Gets the backup out of the SQL Server quickly

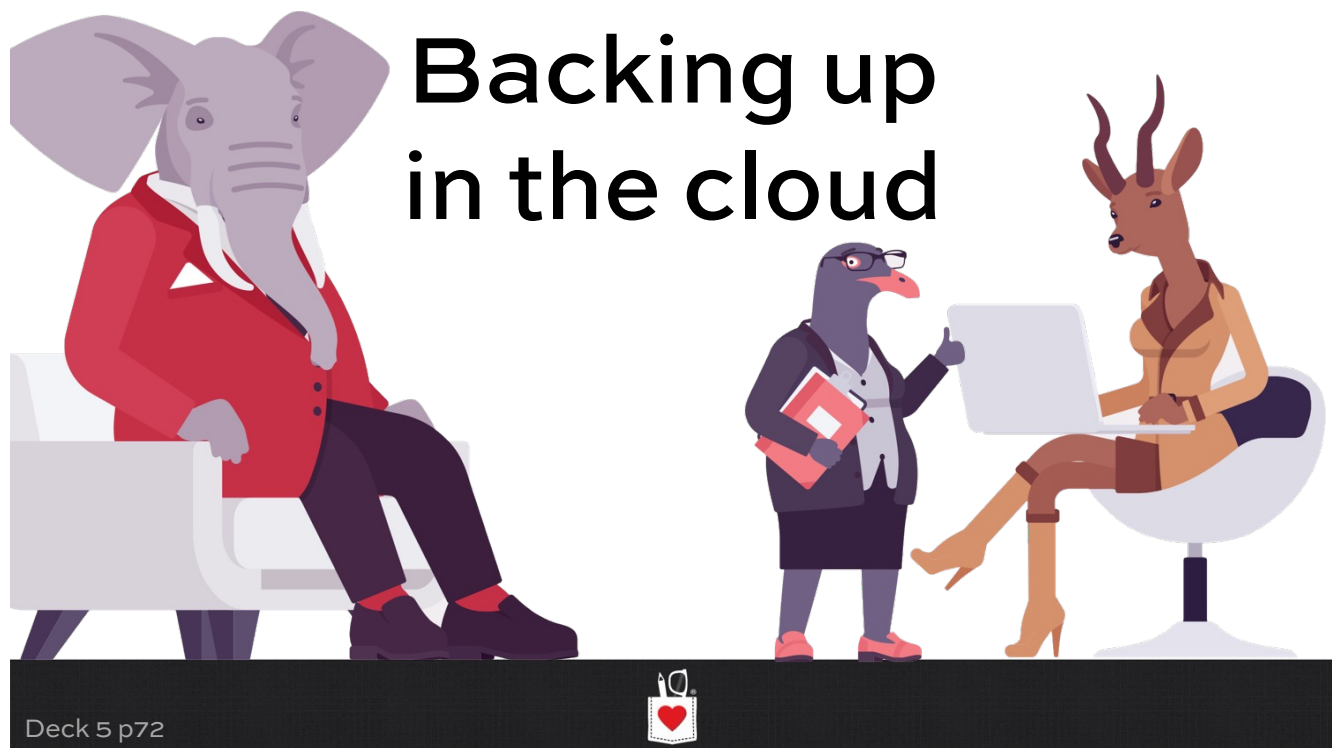
Eliminates the human factor of file sync jobs

Easy for me to restore backups from unrelated servers
(dev, QA, AG replicas)

Easy for sysadmins to configure retention, replication in
AWS S3 & Azure Blob Storage

Easy for sysadmins to sync to other cloud accounts to
avoid account emergencies

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Elliot's Enterprise backups

One backup tool for the entire shop:
still use this, but make sure it does EBS/Blob
snapshots and accesses them directly, not
through the VM

Then do these via snapshot automation:

- Policy-based retention, encryption
- Syncs offsite automatically

DBAs still have minimal control or input

Backups & restores can get faster

Deck 5 p73



Billie's Small Business Backups



Most servers use a VM-based backup tool.
This is still fine – but again, make sure the tool
does EC2/Blob snapshots.

If it doesn't, it's time to switch tools, or switch to
native SQL Server backups.

For native SQL Server backups, write to:

1. Best: cloud-native file storage (S3, Blob)
2. Good: cloud file servers (FSx, Azure Files)

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Alex the Architect's backups

For native SQL Server backups, write to:

1. Best: cloud-native file storage (S3, Blob)
2. Good: cloud file servers (FSx, Azure Files)

Configure that file storage for retention, replication.

Write backups with:

1. Best: a version of SQL that supports it (2022+ for S3, 2012+ for Azure Blobs)
2. Good: a 3rd party backup tool (Idera SQL Safe, Quest Litespeed, Redgate SQL Backup)

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Design backups first.

For a successful cloud migration:

1. Design how you want to back up SQL Server.
2. If your solution requires native backup & restore, not using EBS/Blob snapshots, provision cloud VMs to test backup/restore speed.
3. You'll often find the backup/restore speed dictates the minimum VM instance type & size you can use.

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Actual client summary

You came to me because queries are too slow.

Unbeknownst to you, the DBA team is only doing full backups weekly because it takes >20 hours.

That also means a full restore might take >20 hours.

During the call, you said you weren't okay with that.

You're on the wrong instance type: we need a LOT more horsepower for backups, not to mention queries.

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Design backups first, then...

As long as backups & restores meet your RPO/RTO:

- You've got the right instance type (to get enough storage throughput)
- You just may not have the right size (to get the query performance you want)

Don't overthink that: it's easy to change size later.

Go live, then measure performance & adjust sizes.

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